

Week 1 Exercises

You don't need to submit exercises to me. I will discuss some of them in class, and I will give you a quiz based on these exercises, just to see if you are doing the exercises,

Page and exercise numbers refer to the book (PDF) Applied Discrete Structures by Al Doerr, Ken Levasseur.

Please write your solutions on paper, with a pen.

1.1.3 Exercises on Page 4

1 through 5

1.1.3 Exercises

1. List four elements of each of the following sets:
 - (a) $\{k \in \mathbb{P} \mid k - 1 \text{ is a multiple of } 7\}$
 - (b) $\{x \mid x \text{ is a fruit and its skin is normally eaten}\}$
 - (c) $\{x \in \mathbb{Q} \mid \frac{1}{x} \in \mathbb{Z}\}$
 - (d) $\{2n \mid n \in \mathbb{Z}, n < 0\}$
 - (e) $\{s \mid s = 1 + 2 + \cdots + n \text{ for some } n \in \mathbb{P}\}$
2. List all elements of the following sets:
 - (a) $\{\frac{1}{n} \mid n \in \{3, 4, 5, 6\}\}$
 - (b) $\{\alpha \in \text{the alphabet} \mid \alpha \text{ precedes F}\}$
 - (c) $\{x \in \mathbb{Z} \mid x = x + 1\}$
 - (d) $\{n^2 \mid n = -2, -1, 0, 1, 2\}$
 - (e) $\{n \in \mathbb{P} \mid n \text{ is a factor of } 24 \}$

3. Describe the following sets using set-builder notation.
- (a) $\{5, 7, 9, \dots, 77, 79\}$
 - (b) the rational numbers that are strictly between -1 and 1
 - (c) the even integers
 - (d) $\{-18, -9, 0, 9, 18, 27, \dots\}$
4. Use set-builder notation to describe the following sets:
- (a) $\{1, 2, 3, 4, 5, 6, 7\}$
 - (b) $\{1, 10, 100, 1000, 10000\}$
 - (c) $\{1, 1/2, 1/3, 1/4, 1/5, \dots\}$
 - (d) $\{0\}$
5. Let $A = \{0, 2, 3\}$, $B = \{2, 3\}$, and $C = \{1, 5, 9\}$. Determine which of the following statements are true. Give reasons for your answers.
- (a) $3 \in A$
 - (b) $\{3\} \in A$
 - (c) $\{3\} \subseteq A$
 - (d) $B \subseteq A$
 - (e) $A \subseteq B$
 - (f) $\emptyset \subseteq C$
 - (g) $\emptyset \in A$
 - (h) $A \subseteq A$

1.2.4 Exercises on Page 9
1 through 7

1.2.4 Exercises

1. Let $A = \{0, 2, 3\}$, $B = \{2, 3\}$, $C = \{1, 5, 9\}$, and let the universal set be $U = \{0, 1, 2, \dots, 9\}$. Determine:
 - (a) $A \cap B$
 - (b) $A \cup B$
 - (c) $B \cup A$
 - (d) $A \cup C$
 - (e) $A - B$
 - (f) $B - A$
 - (g) A^c
 - (h) C^c
 - (i) $A \cap C$
 - (j) $A \oplus B$
2. Let A , B , and C be as in Exercise 1, let $D = \{3, 2\}$, and let $E = \{2, 3, 2\}$. Determine which of the following are true. Give reasons for your decisions.
 - (a) $A = B$
 - (b) $B = C$
 - (c) $B = D$
 - (d) $E = D$
 - (e) $A \cap B = B \cap A$
 - (f) $A \cup B = B \cup A$
 - (g) $A - B = B - A$
 - (h) $A \oplus B = B \oplus A$
3. Let $U = \{1, 2, 3, \dots, 9\}$. Give examples of sets A , B , and C for which:
 - (a) $A \cap (B \cap C) = (A \cap B) \cap C$
 - (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
 - (c) $(A \cup B)^c = A^c \cap B^c$
 - (d) $A \cup A^c = U$
 - (e) $A \subseteq A \cup B$
 - (f) $A \cap B \subseteq A$

4. Let $U = \{1, 2, 3, \dots, 9\}$. Give examples to illustrate the following facts:
 - (a) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
 - (b) There are sets A and B such that $A - B \neq B - A$
 - (c) If $U = A \cup B$ and $A \cap B = \emptyset$, it always follows that $A = U - B$.
5. What can you say about A if $U = \{1, 2, 3, 4, 5\}$, $B = \{2, 3\}$, and (separately)
 - (a) $A \cup B = \{1, 2, 3, 4\}$
 - (b) $A \cap B = \{2\}$
 - (c) $A \oplus B = \{3, 4, 5\}$
6. Suppose that U is an infinite universal set, and A and B are infinite subsets of U . Answer the following questions with a brief explanation.
 - (a) Must A^c be finite?
 - (b) Must $A \cup B$ be infinite?
 - (c) Must $A \cap B$ be infinite?
7. Given that U = all students at a university, D = day students, M = mathematics majors, and G = graduate students. Draw Venn diagrams illustrating this situation and shade in the following sets:

(a) evening students	(c) non-math graduate students
(b) undergraduate mathematics majors	(d) non-math undergraduate students